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Introduction

pISSN 2287-2728 eISSN 2287-285X <https://doi.org/10.3350/cmh.2022.0365> Review Clinical and Molecular Hepatology 2023;29(Suppl):S32-S42 Global incidence and prevalence of nonalcoholic fatty liver disease Margaret LP Teng^{1,2,*}, Cheng Han Ng^{2,*}, Daniel Q. Huang^{1,2,3}, Kai En Chan², Darren JH Tan², Wen Hui Lim², Ju Dong Yang⁴, Eunice Tan^{1,2,3}, and Mark D. Muthiah^{1,2,3} 1Division of Gastroenterology and Hepatology, Department of Medicine, National University Hospital, Singapore; 2Yong Loo Lin School of Medicine, National University of Singapore, Singapore; 3National University Centre for Organ Transplantation, National University Hospital, Singapore; 4Karsh Division of Gastroenterology and Hepatology, Cedars-Sinai Medical Center, Los Angeles, CA, USA Nonalcoholic fatty liver disease (NAFLD) is a leading cause of liver disease worldwide. The estimated global incidence of NAFLD is 47 cases per 1,000 population and is higher among males than females. The estimated global prevalence of NAFLD among adults is 32% and is higher among males (40%) compared to females (26%). The global prevalence of NAFLD has increased over time, from 26% in studies from 2005 or earlier to 38% in studies from 2016 or beyond. The prevalence of NAFLD varies substantially by world region, contributed by differing rates of obesity, and genetic and socioeconomic factors. The prevalence of NAFLD exceeds 40% in the Americas and South-East Asia. The prevalence of NAFLD is projected to increase significantly in multiple world regions by 2030 if current trends are left unchecked. In this review, we discuss trends in the global incidence and prevalence of NAFLD and discuss future projections. (Clin Mol Hepatol 2023;29(Suppl):S32-S42) Keywords: Nonalcoholic fatty liver disease; Incidence; Prevalence; Epidemiology INTRODUCTION hepatitis (NASH), which can progress to liver fibrosis and cirrhosis.² The global prevalence of NAFLD has been increasing Nonalcoholic fatty liver disease (NAFLD) is a leading cause over time, with a recent meta-analysis estimating that 32% of of chronic liver disease globally.¹ It encompasses a spectrum the adult population is afflicted by NAFLD.³ This has occurred ranging from simple hepatic steatosis to nonalcoholic steato- in tandem with the global obesity and diabetes epidemics.^{4,5} Corresponding author : Daniel Q. Huang Division of Gastroenterology and Hepatology, Department of Medicine, National University Health System, Tower Block Level 10, 1E Kent Ridge Road, Singapore 119228, Singapore Tel: +65 6772 4354, Fax: +65 6775 1518, E-mail: daniel_huang@nus.edu.sg <https://orcid.org/0000-0002-5165-5061> Mark D. Muthiah Division of Gastroenterology and Hepatology, Department of Medicine, National University Health System, Tower Block Level 10, 1E Kent Ridge Road, Singapore 119228, Singapore Tel: +65 6772 4354, Fax: +65 6775 1518, E-mail: mdcmdm@nus.edu.sg <https://orcid.org/0000-0002-9724-4743> *Margaret LP Teng and Cheng Han Ng contributed equally as co-first authors. Editor: Sang Gyune Kim, Soonchunhyang University Hospital Bucheon, Korea Received : Nov. 2, 2022 / Revised : Dec. 6, 2022 / Accepted : Dec. 12, 2022 Copyright © 2023 by Korean Association for the Study of the Liver This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/3.0/>) which

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Clinical Findings

pISSN 2287-2728 eISSN 2287-285X <https://doi.org/10.3350/cmh.2022.0365> Review Clinical and Molecular Hepatology 2023;29(Suppl):S32-S42 Global incidence and prevalence of nonalcoholic fatty liver disease Margaret LP Teng^{1,2,*}, Cheng Han Ng^{2,*}, Daniel Q. Huang^{1,2,3}, Kai En Chan², Darren JH Tan², Wen Hui Lim², Ju Dong Yang⁴, Eunice Tan^{1,2,3}, and Mark D. Muthiah^{1,2,3} ¹Division of Gastroenterology and Hepatology, Department of Medicine, National University Hospital, Singapore; ²Yong Loo Lin School of Medicine, National University of Singapore, Singapore; ³National University Centre for Organ Transplantation, National University Hospital, Singapore; ⁴Karsh Division of Gastroenterology and Hepatology, Cedars-Sinai Medical Center, Los Angeles, CA, USA Nonalcoholic fatty liver disease (NAFLD) is a leading cause of liver disease worldwide. The estimated global incidence of NAFLD is 47 cases per 1,000 population and is higher among males than females. The estimated global prevalence of NAFLD among adults is 32% and is higher among males (40%) compared to females (26%). The global prevalence of NAFLD has increased over time, from 26% in studies from 2005 or earlier to 38% in studies from 2016 or beyond. The prevalence of NAFLD varies substantially by world region, contributed by differing rates of obesity, and genetic and socioeconomic factors. The prevalence of NAFLD exceeds 40% in the Americas and South-East Asia. The prevalence of NAFLD is projected to increase significantly in multiple world regions by 2030 if current trends are left unchecked. In this review, we discuss trends in the global incidence and prevalence of NAFLD and discuss future projections. (Clin Mol Hepatol 2023;29(Suppl):S32-S42) Keywords: Nonalcoholic fatty liver disease; Incidence; Prevalence; Epidemiology INTRODUCTION hepatitis (NASH), which can progress to liver fibrosis and cirrhosis.² The global prevalence of NAFLD has been increasing Nonalcoholic fatty liver disease (NAFLD) is a leading cause over time, with a recent meta-analysis estimating that 32% of of chronic liver disease globally.¹ It encompasses a spectrum the adult population is afflicted by NAFLD.³ This has occurred ranging from simple hepatic steatosis to nonalcoholic steato- in tandem with the global obesity and diabetes epidemics.^{4,5} Corresponding author : Daniel Q. Huang Division of Gastroenterology and Hepatology, Department of Medicine, National University Health System, Tower Block Level 10, 1E Kent Ridge Road, Singapore 119228, Singapore Tel: +65 6772 4354, Fax: +65 6775 1518, E-mail: daniel_huang@nus.edu.sg <https://orcid.org/0000-0002-5165-5061> Mark D. Muthiah Division of Gastroenterology and Hepatology, Department of Medicine, National University Health System, Tower Block Level 10, 1E Kent Ridge Road, Singapore 119228, Singapore Tel: +65 6772 4354, Fax: +65 6775 1518, E-mail: mdcmdm@nus.edu.sg <https://orcid.org/0000-0002-9724-4743> *Margaret LP Teng and Cheng Han Ng contributed equally as co-first authors. Editor: Sang Gyune Kim, Soonchunhyang University Hospital Bucheon, Korea Received : Nov. 2, 2022 / Revised : Dec. 6, 2022 / Accepted : Dec. 12, 2022 Copyright © 2023 by Korean Association for the Study of the Liver This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/3.0/>) which

permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited. Clinical and Molecular Hepatology Volume_29 Supplement February 2023 also pooled data from 245 studies (2,699,627 individuals) and estimated the global prevalence of NAFLD at 29.8%, which is consistent with Riazi's findings. Likewise, in this study, there was limited or no data from Africa, Oceania, and North and South America. Asia The prevalence of NAFLD varies substantially by region (Fig. 2). The overall prevalence of NAFLD in Asia is approximately 30%. A meta-analysis by Le et al.¹² conducted a literature search in 2019 (182 studies with 2,385,999 individuals) and estimated NAFLD prevalence in Asia at 30.5%. A recent meta- analysis by Riazi et al.³ which included 63 studies (1,000,681 individuals) found that NAFLD prevalence in Asia was 31.6%. This is consistent with a previous meta-analysis by Li et al.¹⁰ which reported NAFLD prevalence in Asia to be 29.62%. The prevalence of NAFLD within Asia is highly variable as it encompasses countries with a wide spectrum of ethnicities and socioeconomic factors. Among Asian subregions, South- east Asia had the highest NAFLD prevalence of 42%. Among Asian countries with more than 3 studies available, Li et al.¹⁰ determined that the highest pooled NAFLD prevalence was in Iran (38.07%), and the lowest pooled NAFLD prevalence was in Japan (22.28%). Riazi et al.³ determined similar results and found that Iran had the highest NAFLD prevalence (40.8%), followed by Taiwan (36.1%), South Korea (34.6%), and China (32.5%). On the other hand, Japan had a strikingly low NAFLD prevalence of 22.3%, which may be related to a low prevalence of obesity. In China, a meta-analysis by Wu et al.¹³ estimated a NAFLD prevalence of 29.88%, and another study by Zhou et al.¹⁴ estimated that NAFLD prevalence was 29.2%. NAFLD prevalence in South Korea is also approximately 30%—a meta-analysis by Im et al.¹⁵ reported a NAFLD prevalence of 30.3%, and Li et al.¹⁰ reported a similar prevalence of 32.9%.¹¹ A large cross- sectional study of 571,872 Korean males in their early 20s found that even among young adult males, NAFLD prevalence was 13.47%, with an increase from 10.66% in 2015 to 16.44% in 2021. There was a higher prevalence of metabolic risk factors such as hypertension, hypercholesterolemia, and hyperglycemia during the same period.¹⁶ Another study utilizing data from Korea National Health and Nutrition Examination Survey found that NAFLD prevalence increased from 18.6% in 1998–2001 to 21.5% in 2016–2017, and there was a S34

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 000,1 Clinical and Molecular Hepatology Volume_29 Supplement February 2023 higher
 prevalence of obesity and diabetes over the same pe- riod.¹⁷ These suggest that the
 increasing NAFLD prevalence may be driven by an increase in metabolic risk factors. Ito et
 al.¹⁸ reported a comparatively lower NAFLD prevalence of 25.5% in Japan, in line with the
 findings by Li and Riazi. This could be attributed to a lower prevalence of obesity and dia-
 betes in Japan compared to other countries,^{19,20} and may be related to a diet that is
 traditionally lower in fat and red meat.²¹ In South Asia, India had a NAFLD prevalence of
 25.7– 32.74%, Bangladesh had a NAFLD prevalence of 26.2–33.86%, and Sri Lanka had a
 NAFLD prevalence of 24.74%.^{3,10} In South- east Asia, Li et al.¹⁰ reported that NAFLD
 prevalence was 38.5% in Malaysia, 40.43% in Singapore, and 51.04% in Indo- nesia. Data
 from Central Asia is lacking, but the Global Bur- den of Disease Study (GBD) 2019 reported
 NAFLD prevalence in Central Asia increased from 12.4% in 1990 to 19.7% in 2019, although
 these estimates require cautious interpretation as the Global Burden of Disease Study relied
 on complex mod- eling and past trends when data was limited.²² A distinct feature of the
 NAFLD epidemic in Asia is the high prevalence of lean NAFLD (body mass index [BMI] <23)
 and non-obese NAFLD (BMI <25).²³ Up to 19% of non-obese Asians have NAFLD,²⁴ which
 may be contributed to a higher percentage of visceral adiposity in Asians compared to other
 ethnicities.²⁵ Visceral adiposity plays an important role in ath- erogenic dyslipidemia and
 insulin resistance. It is a major risk factor for type 2 diabetes and has been implicated in the
 de- velopment and progression of NAFLD.²⁶ Asians also tend to develop diabetes at a
 younger age and lower BMI level, re- sulting in a longer duration of disease and increased
 likeli- hood of complications.²⁷⁻²⁹ Worryingly, emerging data sug- gest that individuals with
 lean NAFLD may be at a higher risk of progressive liver disease, but this hypothesis requires
 vali- dation.³⁰ Europe Meta-analyses by Le and Riazi had similar estimates of the prevalence
 of NAFLD in Europe at 30.9% (11 studies with 15,062 individuals)¹² and 32.6% (7 studies
 with 14,111 individ- uals),³ respectively. Another meta-analysis by Cholongitas et al.³¹
 pooled data from 17 studies (85,203 individuals) and esti- mated NAFLD prevalence in
 Europe to be 26.9%. Cholongitas also found that NAFLD prevalence in Mediterranean coun-
 S36 <https://doi.org/10.3350/cmh.2022.0365> <http://www.e-cmh.org> deunitnoC .2 elbaT
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 ,DLFAN Clinical and Molecular Hepatology Volume_29 Supplement February 2023 tries at
 23.9% compared to non-Mediterranean countries at included in this meta-analysis were
 hospital-based studies 28.5%, although the difference was not statistically signifi- and
 included patients with risk factors for NAFLD, hence the cant. results may not have been fully
 representative of the general Within Europe, Turkey had the highest NAFLD prevalence
 population. Le et al.¹² pooled data from 3 studies (5,716 indi- at 48.4%, followed by Italy at
 38.2%. Germany, Portugal, and viduals) and determined that South America had the greatest
 Spain had NAFLD prevalence between 25–27%.³ A cross-sec- estimated NAFLD prevalence

among the continents at 35.7%. This may be due to a combination of genetic susceptibility and a greater prevalence of metabolic risk factors.⁴⁴ There is a 18.2%.³² A study involving individuals from population-based high prevalence of PNPLA3 genetic polymorphism in the studies in Russia reported that NAFLD prevalence was 40% in general population, especially among individuals with Native the Ural Eye and Medical Study (UEMS) (5,852 individuals), American ancestry.⁴⁵⁻⁴⁷ Furthermore, obesity is extremely and 69.8% in the Ural Very Old Study (UVOS) (1,130 individuals).³³ However, it should be noted that in the UVOS, individual-geographical regions found that central obesity was highest among individuals older with minimum age of 85 years, and methods in South America.⁴⁸ Type 2 diabetes has also been rising in for diagnosis of NAFLD differed between studies as well. prevalence in South America.⁴⁹ Data from the meta-analysis by Le showed that compared to other regions, NAFLD individuals in South America had a higher likelihood of having diabetes and higher mean cholesterol levels.¹² In addition, Based on subgroup data from 4 studies (18,356 individuals), physical activity is often inadequate—Latin America was Le et al.¹² estimated that the prevalence of NAFLD in North ranked as the top region for physical inactivity, with a third of America was 35.3%. More recently, NAFLD prevalence was the population experiencing a lack of physical activity.⁵⁰ reported at 47.8% in the meta-analysis by Riazi which included 2 large studies with 15,178 individuals from the USA.³ This Africa is driven by a high prevalence of obesity in the USA. In North America, Hispanics have the highest NAFLD prevalence, followed by non-Hispanic Whites and non-Hispanic blacks.³⁴⁻³⁹ of NAFLD. A meta-analysis estimated the prevalence of Based on data from National Health and Nutrition Examination Surveys (NHANES) 2017–2018, NAFLD prevalence was estimated at 63.7% in Hispanics, 56.8% in non-Hispanic whites, 28.2% in the meta-analysis by Le, and 56.8% in the meta-analysis by Riazi.^{3,12} Of note, the meta-analysis by Riazi only to genetic factors like the patatin-like phospholipase domain-containing protein 3 (PNPLA3) mutation, which is associated with elevated risk for hepatic steatosis and NASH, occurring more frequently in Hispanics.^{40,41} This could also be related to metabolic factors like the higher prevalence of Oceania central adiposity and insulin resistance in Hispanics compared to non-Hispanic whites.^{34,42} Lower serum triglyceride levels in African-Americans may also contribute to reduced prevalence of NAFLD. Population-based studies NAFLD prevalence.³⁸ using fatty liver index have demonstrated NAFLD prevalence of 35.7–38% in Australia.^{54,55} There are no population-based South America studies on NAFLD prevalence using imaging modalities such as ultrasound.⁵⁶ A meta-analysis comprising 19 studies (5,626 individuals) by Rojas et al. estimated the prevalence of NAFLD in South America at up to 59%.⁴³ Notably, the majority of the studies S38 <https://doi.org/10.3350/cmh.2022.0365> <http://www.e-cmh.org>

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Graphs

Margaret LP Teng, et al. Global incidence and prevalence of NAFLD NASH is now the fastest-rising cause of hepatocellular carcinoma person-years in Japan (Fig. 1).¹⁰ The NAFLD incidence in South Korea worldwide^{6,7} and is also the fastest-rising indication for Korea was around 45 cases per 1,000 person-years.^{10,11} Taken liver transplantation in the United States.⁸ together, the estimates for NAFLD incidence in Asia remain consistent across several meta-analyses (Table 1). INCIDENCE OF NAFLD PREVALENCE OF NAFLD A recent meta-analysis by Riazi et al.³ estimated the incidence of NAFLD at 46.9 cases per 1,000 person-years. The in- Riazi et al.³ pooled data from 72 studies (1,030,160 individuals) of NAFLD was higher in males (70.8 cases per 1,000) and estimated that the global prevalence of NAFLD in person-years) vs. females (26.9 cases per 1,000 person-years, adults was 32% (Table 2). The prevalence was higher in males $P < 0.0001$. However, all included studies were conducted in than females (40% vs. 26%, $P < 0.0001$). The prevalence of Asia, hence it is unclear whether these data are generalizable NAFLD increased from 26% in studies from 2005 or earlier to to other parts of the world. A previous meta-analysis published in 2016 had estimated the NAFLD incidence at 52.34 study by Riazi et al.³ requires cautious interpretation, as data per 1,000 person-years in Asia and 28.01 per 1,000 person-years were available from only 17 countries, hence it is unclear if years in Israel.⁹ Another meta-analysis focused on NAFLD in the estimates from this study are a true reflection of 'global' Asia reported an incidence of 50.9 per 1,000 person-years, prevalence. The relative lack of studies emphasizes the need with the highest incidence of 63 per 1,000 person-years in to improve data collection from regions such as Africa, Oceania, mainland China and the lowest incidence of 29 per 1,000 person-years, and South America, where data was lacking. Le et al.¹² Estimated incidence of nonalcoholic fatty liver disease ≥ 40 cases per 1000 person-years 30-40 cases per 1000 person-years < 30 cases per 1000 person-years Created with mapchart.net Figure 1. Estimated incidence of nonalcoholic fatty liver disease. Data for China, Hong Kong, Japan, and South Korea was obtained from Li et al.¹⁰ and Riazi et al.³. Data for Israel was obtained from Riazi et al.³ and Younossi et al.⁹. NAFLD, nonalcoholic fatty liver disease. Abbreviations: NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; HCC, hepatocellular carcinoma; BMI, body mass index; NHANES, National Health and Nutrition Examination Surveys; PNPLA3, patatin-like phospholipase domain-containing protein 3; FLI, fatty liver index S33 <http://www.e-cmh.org> <https://doi.org/10.3350/cmh.2022.0365> Margaret LP Teng, et al. Global incidence and prevalence of NAFLD Estimated prevalence of nonalcoholic fatty liver disease $\geq 40\%$ 35-39.9% 30-34.9% 25-29.9% 20-25% Created with mapchart.net Figure 2. Estimated prevalence of nonalcoholic fatty liver disease (NAFLD). Data for Iran, China, Taiwan, South Korea, Europe (Turkey, Italy, Germany, Portugal, Spain), North America (USA), and Egypt was obtained from Riazi et al.³. Data for South America was obtained from Le et al.¹². Data for South Asia (India, Bangladesh, Sri Lanka) and Southeast Asia (Indonesia, Malaysia,

Singapore) was obtained from Li et al.¹⁰. Data for Japan was obtained from Li et al.¹⁰ and Riazi et al.³. Data for Sudan was obtained from Younossi et al.⁹. Data for Australia was obtained from population-based studies by Farrell et al.⁵⁵ and Roberts et al.⁵⁴. Projections for prevalence of NAFLD 35 30 25 20 15 10 5 0 China South Korea Singapore Saudi Arabia Japan Germany France Italy Spain UK US Canada Australia 2015-2019 2030 Figure 3. Estimated projections for the prevalence of nonalcoholic fatty liver disease (NAFLD). Data was obtained from Estes et al.^{58,59,61}. Data for Saudi Arabia was obtained from Alswat et al.⁶⁰, data for Canada was obtained from Swain et al.⁶², and data for Australia was obtained from Adams et al.⁶³. S37 <http://www.e-cmh.org> <https://doi.org/10.3350/cmh.2022.0365> Clinical and Molecular Hepatology Volume_29 Supplement February 2023 tries at 23.9% compared to non-Mediterranean countries at included in this meta-analysis were hospital-based studies 28.5%, although the difference was not statistically signifi- and included patients with risk factors for NAFLD, hence the cant. results may not have been fully representative of the general Within Europe, Turkey had the highest NAFLD prevalence population. Le et al.¹² pooled data from 3 studies (5,716 indi- at 48.4%, followed by Italy at 38.2%. Germany, Portugal, and viduals) and determined that South America had the greatest Spain had NAFLD prevalence between 25–27%.³ A cross-sec- estimated NAFLD prevalence among the continents at 35.7%. tional study utilizing data from a large population-based co- This may be due to a combination of genetic susceptibility hort in France found that NAFLD prevalence in France was and a greater prevalence of metabolic risk factors.⁴⁴ There is a 18.2%.³² A study involving individuals from population-based high prevalence of PNPLA3 genetic polymorphism in the studies in Russia reported that NAFLD prevalence was 40% in general population, especially among individuals with Native the Ural Eye and Medical Study (UEMS) (5,852 individuals), American ancestry.⁴⁵⁻⁴⁷ Furthermore, obesity is extremely and 69.8% in the Ural Very Old Study (UVOS) (1,130 individu- common in the region—a cross-sectional study across 4 als).³³ However, it should be noted that in the UVOS, individu- geographical regions found that central obesity was highest als were older with minimum age of 85 years, and methods in South America.⁴⁸ Type 2 diabetes has also been rising in for diagnosis of NAFLD differed between studies as well. prevalence in South America.⁴⁹ Data from the meta-analysis by Le showed that compared to other regions, NAFLD indi- North America viduals in South America had a higher likelihood of having diabetes and higher mean cholesterol levels.¹² In addition, Based on subgroup data from 4 studies (18,356 individuals), physical activity is often inadequate—Latin America was Le et al.¹² estimated that the prevalence of NAFLD in North ranked as the top region for physical inactivity, with a third of America was 35.3%. More recently, NAFLD prevalence was the population experiencing a lack of physical activity.⁵⁰ reported at 47.8% in the meta-analysis by Riazi which includ- ed 2 large studies with 15,178 individuals from the USA.³ This Africa is driven by a high prevalence of obesity in the USA. In North America, Hispanics have the highest NAFLD prevalence, fol- There is a paucity of data from Africa on the epidemiology lowed by non-Hispanic Whites and non-Hispanic blacks.³⁴⁻³⁹ of NAFLD. A meta-analysis estimated the prevalence of Based on data from National Health and Nutrition Examina- NAFLD in Africa at 13.5%, ranging from 9% in Nigeria to 20% tion Surveys (NHANES) 2017–2018, NAFLD prevalence was es- in

Sudan.⁵¹ More recently, NAFLD prevalence was reported at 63.7% in Hispanics, 56.8% in non-Hispanic whites, 28.2% in the meta-analysis by Le, and 56.8% in the meta- and 46.2% in non-Hispanic blacks.³⁷ This could be attributed analysis by Riazi.^{3,12} Of note, the meta-analysis by Riazi only to genetic factors like the patatin-like phospholipase do- included 1 study from Egypt. The wide variation in estimates main-containing protein 3 (PNPLA3) mutation, which is asso- of NAFLD prevalence is likely related to a lack of reliable data ciated with elevated risk for hepatic steatosis and NASH, oc- from Africa.^{52,53} ccurring more frequently in Hispanics.^{40,41} This could also be related to metabolic factors like the higher prevalence of Oceania central adiposity and insulin resistance in Hispanics com- pared to non-Hispanic whites.^{34,42} Lower serum triglyceride Likewise, there is scarce data from Oceania on the inci- levels in African-Americans may also contribute to reduced dence and prevalence of NAFLD. Population-based studies NAFLD prevalence.³⁸ using fatty liver index have demonstrated NAFLD prevalence of 35.7–38% in Australia.^{54,55} There are no population-based South America studies on NAFLD prevalence using imaging modalities such as ultrasound.⁵⁶ A meta-analysis comprising 19 studies (5,626 individuals) by Rojas et al. estimated the prevalence of NAFLD in South America at up to 59%.⁴³ Notably, the majority of the studies S38 <https://doi.org/10.3350/cmh.2022.0365> <http://www.e-cmh.org> Margaret LP Teng, et al. Global incidence and prevalence of NAFLD Adiposity and Its Relationship With Cardiometabolic Risk/Intra- transient elastography among US adults: results from NHANES Abdominal Adiposity. *Am J Clin Nutr* 2012;96:714-726. 2017-2018. *PLoS One* 2021;16:e0252164. 26. Hanlon CL, Yuan L. Nonalcoholic fatty liver disease: the role of 38. Sherif ZA, Saeed A, Ghavimi S, Nouraie SM, Laiyemo AO, Brim visceral adipose tissue. *Clin Liver Dis (Hoboken)* 2022;19:106- H, et al. Global epidemiology of nonalcoholic fatty liver dis- 110. ease and perspectives on US minority populations. *Dig Dis Sci* 27. Yoon KH, Lee JH, Kim JW, Cho JH, Choi YH, Ko SH, et al. Epidemic 2016;61:1214-1225. obesity and type 2 diabetes in Asia. *Lancet* 2006;368:1681-1688. 39. Arshad T, Golabi P, Henry L, Younossi ZM. Epidemiology of non- 28. Huxley R, James WP, Barzi F, Patel JV, Lear SA, Suriyawongpaisal alcoholic fatty liver disease in North America. *Curr Pharm Des P*, et al. Ethnic comparisons of the cross-sectional relationships 2020;26:993-997. between measures of body size with diabetes and hyperten- 40. Romeo S, Kozlitina J, Xing C, Pertsemlidis A, Cox D, Pennacchio sion. *Obes Rev* 2008;9 Suppl 1:53-61. LA, et al. Genetic variation in PNPLA3 confers susceptibility to 29. Chan JC, Malik V, Jia W, Kadowaki T, Yajnik CS, Yoon KH, et al. nonalcoholic fatty liver disease. *Nat Genet* 2008;40:1461-1465. Diabetes in Asia: epidemiology, risk factors, and pathophysiol- 41. Wagenknecht LE, Palmer ND, Bowden DW, Rotter JI, Norris JM, ogy. *JAMA* 2009;301:2129-2140. Ziegler J, et al. Association of PNPLA3 with non-alcoholic fatty 30. Hagström H, Nasr P, Ekstedt M, Hammar U, Stål P, Hultcrantz R, liver disease in a minority cohort: the Insulin Resistance Athero- et al. Risk for development of severe liver disease in lean pa- scleriosis Family Study. *Liver Int* 2011;31:412-416. tients with nonalcoholic fatty liver disease: a long-term follow- 42. Guerrero R, Vega GL, Grundy SM, Browning JD. Ethnic differ- up study. *Hepatol Commun* 2018;2:48-57. ences in hepatic steatosis: an insulin resistance paradox? Hepa- 31. Cholongitas E, Pavlopoulou I, Papatheodoridi M, Markakis GE, tology 2009;49:791-801. Bouras E, Haidich AB, et al.

Epidemiology of nonalcoholic fatty 43. Rojas YAO, Cuellar CLV, Barrón KMA, Arab JP, Miranda AL. Non- liver disease in Europe: a systematic review and meta-analysis. alcoholic fatty liver disease prevalence in Latin America: a sys- Ann Gastroenterol 2021;34:404-414. tematic review and meta-analysis. Ann Hepatol 2022;27:100706. 32. Nabi O, Lacombe K, Boursier J, Mathurin P, Zins M, Serfaty L. 44. Pinto Marques Souza de Oliveira C, Pinchemel Cotrim H, Arrese Prevalence and risk factors of nonalcoholic fatty liver disease M. Nonalcoholic fatty liver disease risk factors in Latin American and advanced fibrosis in general population: the French Na- populations: current scenario and perspectives. Clin Liver Dis tionwide NASH-CO Study. Gastroenterology 2020;159:791-793. (Hoboken) 2019;13:39-42. e2. 45. Arrese M, Arab JP, Riquelme A, Benítez CE, Barrera F, Soza A, 33. Bikbov MM, Gilmanshin TR, Zainullin RM, Kazakbaeva GM, Iaku- et al. High prevalence of PNPLA3 rs738409 (I148M) polymor- pova EM, Fakhretdinova AA, et al. Prevalence of non-alcoholic phism in Chilean latino population and its association to non- fatty liver disease in the Russian Ural Eye and Medical Study alcoholic fatty liver disease risk and histological disease sever- and the Ural Very Old Study. Sci Rep 2022;12:7842. ity. Hepatology 2015;62 Suppl 1:1285A. 34. Browning JD, Szczepaniak LS, Dobbins R, Nuremberg P, Horton 46. Pontoriero AC, Trinks J, Hulaniuk ML, Caputo M, Fortuny L, Pratx JD, Cohen JC, et al. Prevalence of hepatic steatosis in an urban LB, et al. Influence of ethnicity on the distribution of genetic population in the United States: impact of ethnicity. Hepatol- polymorphisms associated with risk of chronic liver disease in ogy 2004;40:1387-1395. South American populations. BMC Genet 2015;16:93. 35. Lazo M, Hernaez R, Eberhardt MS, Bonekamp S, Kamel I, Gual- 47. Chinchilla-López P, Ramírez-Pérez O, Cruz-Ramón V, Canizales- lar E, et al. Prevalence of nonalcoholic fatty liver disease in the Quinteros S, Domínguez-López A, Ponciano-Rodríguez G, et al. United States: the Third National Health and Nutrition Examina- More evidence for the genetic susceptibility of Mexican popu- tion Survey, 1988-1994. Am J Epidemiol 2013;178:38-45. lation to nonalcoholic fatty liver disease through PNPLA3. Ann 36. Kim D, Kim W, Adejumo AC, Cholanteril G, Tighe SP, Wong RJ, Hepatol 2018;17:250-255. et al. Race/ethnicity-based temporal changes in prevalence of 48. Patel SA, Ali MK, Alam D, Yan LL, Levitt NS, Bernabe-Ortiz A, et NAFLD-related advanced fibrosis in the United States, 2005- al. Obesity and its relation with diabetes and hypertension: a 2016. Hepatol Int 2019;13:205-213. cross-sectional study across 4 geographical regions. Glob Heart 37. Zhang X, Heredia NI, Balakrishnan M, Thrift AP. Prevalence and 2016;11:71-79.e4. factors associated with NAFLD detected by vibration controlled 49. Gallardo-Rincón H, Cantoral A, Arrieta A, Espinal C, Magnus MH, S41 <http://www.e-cmh.org> <https://doi.org/10.3350/cmh.2022.0365>

Summary

The provided medical content details the global and regional prevalence, incidence, and projected increases of Non-Alcoholic Fatty Liver Disease (NAFLD). A comprehensive analysis of 245 studies, encompassing 2,699,627 individuals, estimated the global NAFLD prevalence at 29.8%. However, some global analyses noted limitations due to scarce data from Africa, Oceania, and parts of North and South America. In Asia, the overall NAFLD prevalence is approximately 30%, with specific meta-analyses reporting figures ranging from 29.62% to 31.6%. The incidence of NAFLD in Asia was estimated at 52.34 per 1,000 person-years in one meta-analysis and 50.9 per 1,000 person-years in another. A notable demographic finding in Asia indicated a higher incidence in males (36.3 cases per 1,000 person-years) compared to females (26.9 cases per 1,000 person-years, $P < 0.0001$). Globally, the prevalence in adults was 32%, with males exhibiting a higher prevalence (40%) than females (26%, $P < 0.0001$). This global adult prevalence also showed an increase from 26% in studies from 2005 or earlier to 38% in studies from 2016 or beyond, although this specific data set was limited to 17 countries. Significant projections for NAFLD prevalence by 2030 highlight a growing burden. In the Middle East, Saudi Arabia is projected to see cases increase from 8.45 million in 2017 to 12.53 million, reaching a 31.7% prevalence. The United Arab Emirates is expected to rise from 0.255 million to 0.372 million cases, with a 30.2% prevalence. European countries also face substantial increases: France from 13.98 million to 16.05 million cases, Germany from 18.45 million to 20.95 million, Italy from 15.22 million to 17.42 million, Spain from 10.53 million to 12.65 million, and the United Kingdom from 14.08 million to 16.92 million by 2030. In South America, a meta-analysis estimated NAFLD prevalence up to 59%, while Australia reported a prevalence of 35.7–38%. The content emphasizes the need for more population-based studies using imaging modalities in regions like South America and Australia.